ESSENTIALLY FINITE TYPE ALGEBRAS OVER A P-RING ARE P-RINGS

THE STACKS PROJECT AUTHORS

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Editorial note. Benign-screen note: the proof contains the phrase 'it is clear that' / 'is obvious' once, used as a legitimate step inside a complete argument; the author proves the statement, so this is not a hint-only proof.

Allowed standard prerequisites (editorial).

- Being a P-ring is local on the ring, and it suffices to check the formal fibres at maximal ideals. Lemma \ref{lemma-check-P-ring-maximal-ideals} and Lemma \ref{lemma-P-local}, cited at proof lines 13015 and 13032-13033. Reduce the claim to a statement about formal fibres.
- Ascent of the P-ring property along quasi-finite maps, and the P-ring property of polynomial rings. Lemma \ref{lemma-P-ring-goes-up-quasi-finite}, cited at proof lines 13010-13011. Reduce to the polynomial algebra case.
- Complete Noetherian rings are G-rings, and finite type algebras over G-rings are G-rings. Propositions \ref{proposition-Noetherian-complete-G-ring} and \ref{proposition-finite-type-over-G-ring}, cited at proof lines 13036-13038. Make the right vertical arrow of the comparison diagram regular.
- The permanence conditions (A), (C) and (D) for property P. used at proof lines 13029-13031, 13039-13041, and 13042-13044. Carry the fibrewise property along the two arrows of the diagram.

Source reference. Original lemma 0BIV: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 14503–14503. The lemma's own canonical tag, quoted so that readers can resolve the label inside the delivered text; it points at the counted statement itself and carries no proof content.

Date: stacks-project:more-algebra.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 0BIT: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 14501–14501. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 0BIU: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 14502–14502. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 0BK8: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 14551–14551. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 07PS: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 9807–9807. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 07PV: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 9810–9810. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Proposition 0.2. *Let R be a P-ring where P satisfies (A), (B), (C), and (D). If $R \to S$ is essentially of finite type then S is a P-ring.*

Proof. Since being a P-ring is a property of the local rings it is clear that a localization of a P-ring is a P-ring. Conversely, if every localization at a prime is a P-ring, then the ring is a P-ring. Thus it suffices to show that $S_{\mathfrak{q}}$ is a P-ring for every finite type R-algebra S and every prime $\mathfrak{q}$ of S. Writing S as a quotient of $R[x_1, \ldots, x_n]$ we see from Lemma 0BIT that it suffices to prove that $R[x_1, \ldots, x_n]$ is a P-ring. By induction on n it suffices to prove that $R[x]$ is a P-ring. Let $\mathfrak{q} \subset R[x]$ be a maximal ideal. By Lemma 0BIU it suffices to show that the fibres of

$$R[x]_{\mathfrak{q}} \longrightarrow R[x]^\wedge_{\mathfrak{q}}$$

have P. If $\mathfrak{q}$ lies over $\mathfrak{p} \subset R$, then we may replace R by $R_{\mathfrak{p}}$. Hence we may assume that R is a Noetherian local P-ring with maximal ideal $\mathfrak{m}$ and that $\mathfrak{q} \subset R[x]$ lies over $\mathfrak{m}$. Note that there is a unique prime $\mathfrak{q}' \subset R^\wedge[x]$ lying over $\mathfrak{q}$. Consider the diagram

$$\begin{array}{ccc} R[x]^\wedge_{\mathfrak{q}} & \longrightarrow & (R^\wedge[x]_{\mathfrak{q}'})^\wedge \\ \uparrow & & \uparrow \\ R[x]_{\mathfrak{q}} & \longrightarrow & R^\wedge[x]_{\mathfrak{q}'} \end{array}$$

Since R is a P-ring the fibres of $R[x] \to R^\wedge[x]$ have P because they are base changes of the fibres of $R \to R^\wedge$ by a finitely generated field extension so (A) applies. Hence the fibres of the lower horizontal arrow have P for example by Lemma 0BK8. The right vertical arrow is regular because $R^\wedge$ is a G-ring (Propositions 07PS and 07PV). It follows that the fibres of the composition $R[x]_{\mathfrak{q}} \to (R^\wedge[x]_{\mathfrak{q}'})^\wedge$ have P by (C). Hence the fibres of the left vertical arrow have P by (D) and the proof is complete. □